

Name **Suggested Solutions** Class: 17S _____ Reg Number: _____



MERIDIAN JUNIOR COLLEGE
JC 2 Preliminary Examination
Higher 2

Chemistry

9729/03

Paper 3 Free Response

13 September 2018

2 hour

Additional Materials: *Data Booklet*
 Writing Paper

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

Answer all questions in Section A and one question from Section B.

Begin each question on a fresh page of writing paper.

Fasten the writing papers behind the given **Cover Page for Questions 1 & 2** and **Cover Page for Questions 3 & 4 or 5** respectively.

Hand in Questions **1 & 2** and **3 & 4 or 5 separately**.

You are advised to spend about 30 minutes per question only.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This document consists of **15** printed pages (including this cover page).

Section A

Answer **all** questions in this section.

- 1(a)** Explain the variation in melting point of the oxides of the elements in Period 3 (sodium to sulfur), using oxides of three different structure as examples.

Indicate clearly the type of structure and bonding present in the examples you have selected.

[3]

High melting point for Na_2O , MgO (or Al_2O_3)

– giant ionic lattice structure with strong electrostatic forces of attraction between oppositely charged ions

High melting point (intermediate between that of Na_2O and MgO) for SiO_2

– giant (3D) molecular structure with strong covalent bonds between (Si and O) atoms

Low melting point (lower than that of Na_2O) for P_4O_{10} , SO_3

– simple molecular structure with weak instantaneous dipole – induced dipole attractive forces between molecules.

- (b)** With the aid of the *Data Booklet*, explain the difference in observed melting points of NaCl (801 °C) and MgO (2852 °C)

[2]

$$\text{Lattice energy} \propto \frac{q_+ \times q_-}{r_+ + r_-}$$

Charge of cation, q_+ : $\text{Na}^+ < \text{Mg}^{2+}$

Radius of cation, r_+ : $\text{Na}^+ (0.095 \text{ nm}) > \text{Mg}^{2+} (0.065 \text{ nm})$

Charge of anion, q_- : $\text{Cl}^- < \text{O}^{2-}$

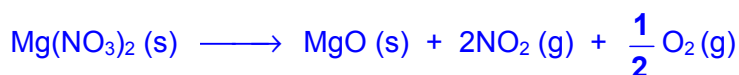
Radius of anion, r_- : $\text{Cl}^- (0.181 \text{ nm}) > \text{O}^{2-} (0.140 \text{ nm})$

Lattice energy of NaCl is less exothermic than that of $\text{MgO} \Rightarrow$ Melting point: $\text{NaCl} < \text{MgO}$

- (c)** Instead of melting like chlorides or oxides, Group 2 nitrates decompose upon heating.

- (i)** Write an equation, including state symbols, for the thermal decomposition of magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$.

[1]



- (ii)** Predict, with reasoning, whether barium nitrate, $\text{Ba}(\text{NO}_3)_2$, is more or less thermally stable compared with $\text{Mg}(\text{NO}_3)_2$.

[2]

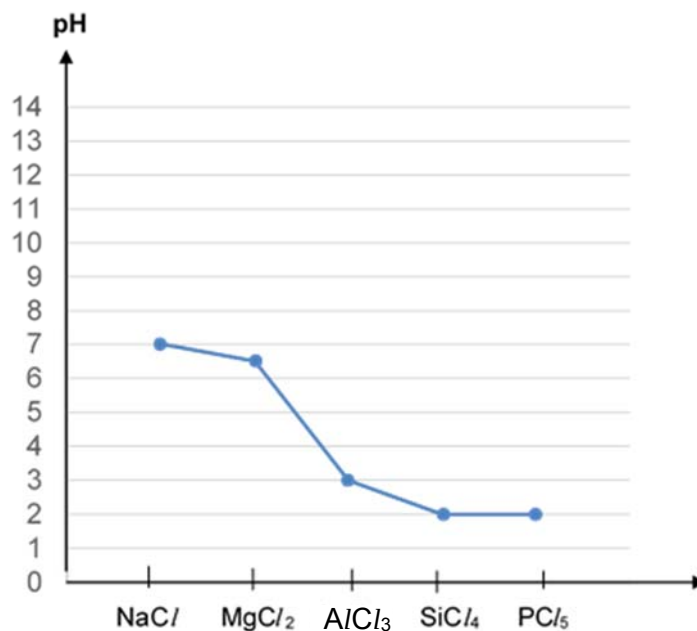
$\text{Ba}(\text{NO}_3)_2$ is more thermally stable compared with $\text{Mg}(\text{NO}_3)_2$.

Ba^{2+} is less polarising due to lower charge density than Mg^{2+} , hence it is less able to polarise electron cloud of NO_3^- to a significant extent.

(d) Chlorides of Period 3 elements dissolve in water to form solutions of differing pH values.

- (i) Sketch a graph to illustrate how the pH of the resultant solutions vary across the period (from sodium to phosphorus).

[1]



- (ii) Write an equation to illustrate the reaction of PCl₅ with water.

[1]

- (iii) Lithium is a Group 1 element. Its chloride, LiCl however behaves similarly to that of a chloride of a Group 2 element when dissolved in water.

Suggest a reason why this would be so.

[1]

Similar to MgCl₂ as the metal cations have similar charge densities (hence diagonal relationship)

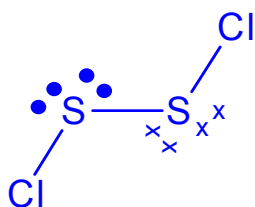
- (e) Disulfur dichloride, S_2Cl_2 is one of three chlorides of sulfur. It is a liquid at room temperature with a pungent odour. It has a structure in which the central sulfur atoms are bonded by a single covalent bond.

Draw clear diagrams of two possible molecular arrangements of S_2Cl_2 , showing clearly the electron pairs around the sulfur atoms. Apply the principles of the VSEPR theory to suggest which arrangement would result in a molecule which could be more stable.

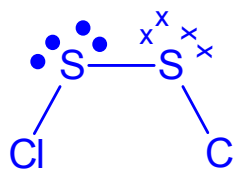
[3]

To minimise repulsion and maximise stability, the 4 electron pairs are directed to corners of a regular tetrahedron. Since lone pair – lone pair repulsion > lone pair – bond pair repulsion > bond pair – bond pair repulsion, the shape is bent about each S atom.

Arrangement 1 is more stable over arrangement 2 as the lone pairs of electrons are pointed away from each other (in opposite directions), which minimises the repulsion.

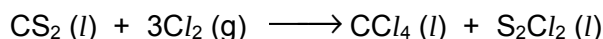


Arrangement 1



Arrangement 2

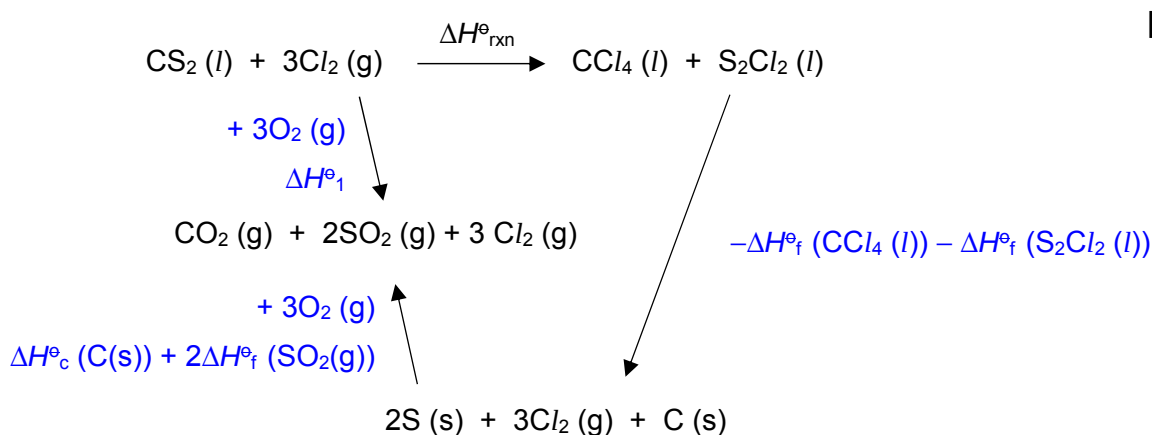
- (f) CCl_4 , an important commercial solvent is prepared by the reaction of Cl_2 gas with a sulfur-containing carbon compound. S_2Cl_2 is produced as a by-product. The reaction for the production is shown below.



- (i) Using appropriate data in **Table 1.2** and the energy cycle provided, determine the enthalpy change, $\Delta H^\circ_{\text{rxn}}$ for the above reaction.

Table 1.2

	$\Delta H^\circ / \text{kJ mol}^{-1}$
Standard enthalpy change of formation of $\text{CCl}_4(l)$	-135.4
Standard enthalpy change of combustion of C (s)	-393.5
Standard enthalpy change of formation of $\text{S}_2\text{Cl}_2(l)$	-58.2
Standard enthalpy change of formation of $\text{SO}_2(g)$	-296.8
$\text{CS}_2(l) + 3\text{O}_2(g) \longrightarrow \text{CO}_2(g) + 2\text{SO}_2(g)$	-1077.0
$\text{SO}_2(g) + \text{Cl}_2(g) \longrightarrow \text{SO}_2\text{Cl}_2(l)$	+97.3



By Hess' Law,

$$\begin{aligned}\Delta H^\circ_{\text{rxn}} &= \Delta H^\circ_1 - \Delta H^\circ_c(\text{C(s)}) - 2\Delta H^\circ_f(\text{SO}_2(\text{g})) - \Delta H^\circ_f(\text{CCl}_4(l)) - \Delta H^\circ_f(\text{S}_2\text{Cl}_2(l)) \\ \Delta H^\circ_{\text{rxn}} &= (-1077) - (-393.5) - 2(-296.8) + (-135.4) + (-58.2) \\ &= -283.5 \text{ kJ mol}^{-1}\end{aligned}$$

- (ii) CCl_4 could be reacted with limited oxygen according to the following reaction.



Predict the sign of entropy change, ΔS° for this reaction and explain whether the reaction would be spontaneous at all temperatures.

[2]

ΔS is positive as there is an increase in number of moles of gas (from 1 to 2), hence an increase in the entropy / disorder of the system.

$$\Delta G = \Delta H - T\Delta S$$

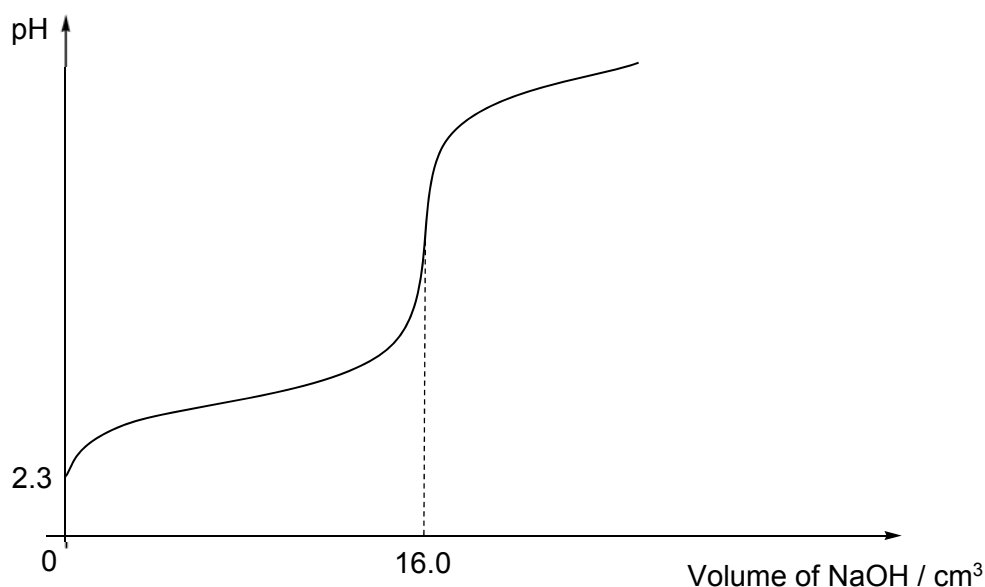
When $\Delta S = +ve$, then $-T\Delta S = -ve$, $\Rightarrow$ Both ΔH and $-T\Delta S$ are negative

$\Delta G < 0$ ($-ve$) $\Rightarrow$ spontaneous for all temperature.

[Total: 18]

2 In analytical chemistry, both quantitative and qualitative information about chemical reactions are established through the use of various chemical methods.

- (a) Glycolic acid, HOCH_2COOH is a weak acid used in many cosmetic skin-care products. When a student titrated 25.0 cm^3 of a glycolic acid against $0.250 \text{ mol dm}^{-3}$ NaOH, the following titration curve was obtained.



- (i) Suggest an appropriate indicator for the titration, giving a reason for your choice.

[1]

Phenolphthalein

The pH transition range of the indicator lies within the rapid pH change over the equivalence point

- (ii) Use the above titration data to calculate the K_a value of glycolic acid.

[2]

$\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$

Amount of NaOH = Amount of HA in 25.0 cm^3 of solution

$$= \frac{16.0}{1000} \times 0.250 = 4.00 \times 10^{-3} \text{ mol}$$

$$[\text{HA}] = \frac{4.00 \times 10^{-3}}{\frac{25.0}{1000}} = 0.160 \text{ mol dm}^{-3}$$

$$[\text{H}^+] = 10^{-2.3} = 5.01 \times 10^{-3} \text{ mol dm}^{-3}$$

$$K_a = \frac{(5.01 \times 10^{-3})^2}{0.160} = 1.57 \times 10^{-4} \text{ mol dm}^{-3}$$

- (b) The pH level of skin-care products need to be regulated to prevent degradation of the active ingredients. A buffer solution containing ethanoic acid and its sodium salt, sodium ethanoate can be added to control the pH. The pK_a of ethanoic acid is 4.76.

- (i) With the aid of a balanced equation, briefly explain how a solution of ethanoic acid and sodium ethanoate can maintain a fairly constant pH when a small amount of acid is added to this solution.

[1]

When a small amount of H^+ is added, $CH_3COO^- + H^+ \longrightarrow CH_3COOH$

H^+ added is removed as CH_3COOH . $[H^+]$ slightly changed and pH remains fairly constant

- (ii) Determine the resultant pH of a 100.0 cm^3 solution containing 0.10 mol dm^{-3} of ethanoic acid and 0.10 mol dm^{-3} of sodium ethanoate when $5.0 \times 10^{-4} \text{ mol}$ of H_2SO_4 is added to this solution.

[2]

$H_2SO_4 \equiv 2H^+$

Amount of H^+ added = $1.0 \times 10^{-3} \text{ mol}$

$[H^+] = 1.0 \times 10^{-2} \text{ mol dm}^{-3}$

$[CH_3COOH]_{\text{new}} = 0.10 + 0.010 = 0.110 \text{ mol dm}^{-3}$

$[CH_3COO^-]_{\text{new}} = 0.10 - 0.010 = 0.0900 \text{ mol dm}^{-3}$

$$pH = pK_a + \log_{10} \frac{[CH_3COO^-]_{\text{new}}}{[CH_3COOH]_{\text{new}}} = 4.76 + \log_{10} \left(\frac{0.0900}{0.110} \right) = 4.67 \text{ (2 d.p.)}$$

- (c) Another student performed a series of thermometric experiments to determine the enthalpy change of neutralisation involving several acids and sodium hydroxide. Her experimental results are summarised in the table below.

Acid	Enthalpy change of neutralisation, $\Delta H_n / \text{kJ mol}^{-1}$
hydrochloric acid, HCl	- 57.2
hydrofluoric acid, HF	- 68.6
glycolic acid, HOCH_2COOH	- 53.2

- (i) Explain why the neutralisation reaction involving glycolic acid is less exothermic than that involving hydrochloric acid.

[1]

Glycolic acid is a weak acid and is only slightly dissociated in aqueous solution. Some of the energy evolved from the neutralisation process is used to further dissociate the weak acid completely.

Deducing the strength of acids solely from the enthalpy change of neutralisation may not be appropriate as there are exceptions. For instance, the neutralisation reaction between HF, a weak acid and NaOH is significantly more exothermic than that of HCl and NaOH.

- (ii) Suggest why the neutralisation reaction between HF and NaOH is unexpectedly more exothermic.

[1]

On dissociation, the small F^- ions can form strong ion–dipole interactions with water molecules that releases high amount of heat.

OR

The enthalpy change of hydration of F^- ions is very exothermic.

- (d) Argentometric titration can be used to determine the amount of sodium chloride present in skin–care products by titrating the sample against silver nitrate, $AgNO_3$. A suitable indicator is one that forms a precipitate of a different colour after all the chloride ions have reacted.

The following are relevant K_{sp} values of some sparingly soluble silver compounds.

	K_{sp}
AgCl	$1.8 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$
AgI	$7.7 \times 10^{-17} \text{ mol}^2 \text{ dm}^{-6}$
Ag_3PO_4	$1.3 \times 10^{-20} \text{ mol}^4 \text{ dm}^{-12}$

- (i) Write an expression for the solubility product, K_{sp} of Ag_3PO_4 .

[1]

$$K_{sp} \text{ of } Ag_3PO_4 = [Ag^+]^3 [PO_4^{3-}]$$

- (ii) Given that Ag_3PO_4 precipitate is yellow in appearance, by means of suitable calculations, suggest whether potassium phosphate, K_3PO_4 can be used as an indicator in the argentometric titration for the determination of chloride content.

[2]

Let the solubility of AgCl be s .



$$K_{sp}(AgCl) = [Ag^+][Cl^-] = s^2$$

$$\text{Solubility of } AgCl = [Ag^+] = \sqrt{1.8 \times 10^{-10}} = 1.34 \times 10^{-5} \text{ mol dm}^{-3}$$

Let the solubility of Ag_3PO_4 be s .



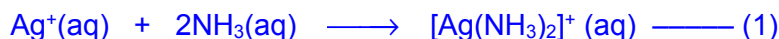
$$K_{sp}(Ag_3PO_4) = [Ag^+]^3 [PO_4^{3-}] = (3s)^3 (s) = 27s^4$$

$$\text{Solubility of } Ag_3PO_4 = \sqrt[4]{\frac{1.3 \times 10^{-20}}{27}} = 4.68 \times 10^{-6} \text{ mol dm}^{-3}$$

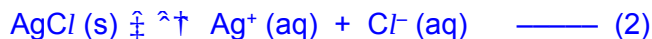
Since solubility of Ag_3PO_4 is lower, it has the tendency to precipitate out first before most Cl^- has been precipitated therefore it is not suitable to be used as an indicator.

- (iii) With the aid of appropriate equations, explain why silver chloride, AgCl is soluble in excess aqueous ammonia. [3]

When aqueous ammonia is added, a soluble complex cation $[\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$ is formed, hence decreasing $[\text{Ag}^+(\text{aq})]$.



By Le Chatelier's Principle, the equilibrium position in (2) shifts right to increase $[\text{Ag}^+(\text{aq})]$ but Ag^+ will be used to form more $[\text{Ag}(\text{NH}_3)_2]^+$.



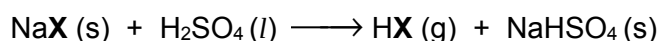
Overall, $[\text{Ag}^+(\text{aq})]$ decreases and ionic product of $[\text{Ag}^+][\text{Cl}^-]$ decreases to a value lower than its relatively higher K_{sp} . Hence white ppt of AgCl dissolves.

- (iv) Hence, briefly explain why silver iodide, AgI, remains insoluble even in concentrated ammonia solution. [1]

K_{sp} of AgI is relatively much lower than that of AgCl. AgI remains insoluble in concentrated NH_3 as its ionic product, $[\text{Ag}^+][\text{I}^-]$ which decreases will still be higher than its K_{sp} value.

- (e) The distinctive reactions of solid halides with hot concentrated sulfuric acid also provide useful basis for further analysis.

These halides react with hot concentrated sulfuric acid to form white fumes of hydrogen halides.



where $\text{X} = \text{Cl}, \text{Br}$ or I

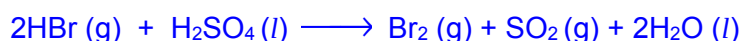
However, only HBr and HI formed will further react with concentrated H_2SO_4 . The main products for these reactions are recorded in the table below.

HCl	HBr	HI
no further reaction	red-brown gas	violet fumes
	acidic gas that decolourises acidified purple KMnO_4	pungent H_2S gas

- (i) Suggest an identity for both the red-brown gas and the acidic gas formed when HBr further reacts with concentrated H_2SO_4 . [1]

Red-brown gas is Br_2 and acidic gas is SO_2 .

- (ii) Hence, write an equation for the reaction between HBr and concentrated H_2SO_4 . [1]



- (iii) By considering the reactions of HCl and HI with concentrated sulfuric acid and the change in oxidation number of sulfur, explain which halide ion is a stronger reducing agent.

[2]

Cl^- is a weaker reducing agent OR I^- is a stronger reducing agent.

Cl^- does not react with H_2SO_4 so there is no change in oxidation number of sulfur.

I^- is able to reduce H_2SO_4 to H_2S . There is a decrease in the oxidation number of S from +6 in H_2SO_4 to -2 in H_2S .

[Total: 19]

- 3 Transition elements show typical properties that distinguish them from s-block elements such as calcium. The following table gives data about some physical properties of the elements calcium, iron, copper and cadmium.

property	calcium	iron	copper	cadmium
relative atomic mass	40.1	55.8	63.5	112.4
atomic radius (metallic) / nm	0.197	0.126	0.128	0.151
ionic radius (2+) / nm	0.099	0.076	0.069	0.097
melting point / °C	839	1535	1085	322
density / g cm ⁻³	1.54	7.86	8.92	8.65

- (a) (i) Explain why the atomic radii of iron and copper are similar to each other.

[2]

- Nuclear charge increases from Fe to Cu. The electrons in Cu are added to the inner 3d orbitals and provide more shielding for the 4s electrons.
- Increase in nuclear charge slightly outweighs the increase in shielding effect. Effective nuclear charge increases only slightly.
- Hence, atomic radius of Fe and Cu are similar.

- (ii) The melting point of iron is significantly higher than that of calcium and cadmium. Explain this using relevant data from the table and in terms of the structure and bonding in each metal.

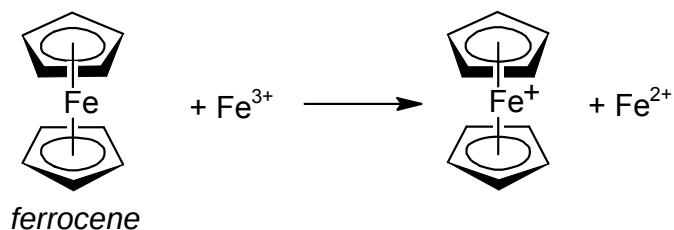
[3]

Fe, Cd and Ca have giant metallic structure. Both 3d and 4s electrons of Fe and Cd are involved in the delocalized in metallic bonding while only the 4s electrons for Ca is involved.

The ionic radius of Fe²⁺ is also smaller than that of Ca²⁺ and Cd²⁺ resulting in higher charge density. This results in stronger electrostatic forces of attraction between the Fe²⁺ cations and sea of delocalised electrons.

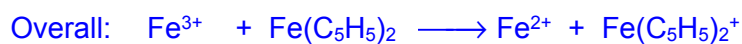
Hence, Fe has a higher melting point than Ca and Cd.

- (b) Ferrocene, $\text{Fe}(\text{C}_5\text{H}_5)_2$, is an orange organometallic solid. In this complex, C_5H_5^- is the ligand and it donates π electrons from the ring to the vacant $3d$ orbitals of Fe. A chemist suggested that ferrocene would react with aqueous iron(III) ions to form iron(II) ions as shown below.



- (i) The chemist found the reduction potential ferrocene, $E^\circ_{\text{Fe}(\text{C}_5\text{H}_5)_2^+ / \text{Fe}(\text{C}_5\text{H}_5)_2}$ to be +0.63V. Predict if the reaction between ferrocene and iron(III) ions will occur. Calculate ΔG° for this reaction.

(ii) [2]



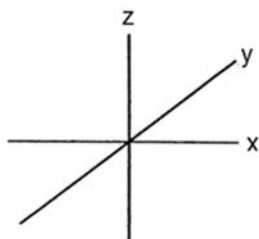
$$\begin{aligned} E^\circ_{\text{cell}} &= E^\circ_{\text{red}} - E^\circ_{\text{oxid}} \\ &= 0.77 - 0.63 \\ &= +0.14 \text{ V} \end{aligned}$$

Since $E^\circ_{\text{cell}} > 0 \Rightarrow$ Reaction is spontaneous

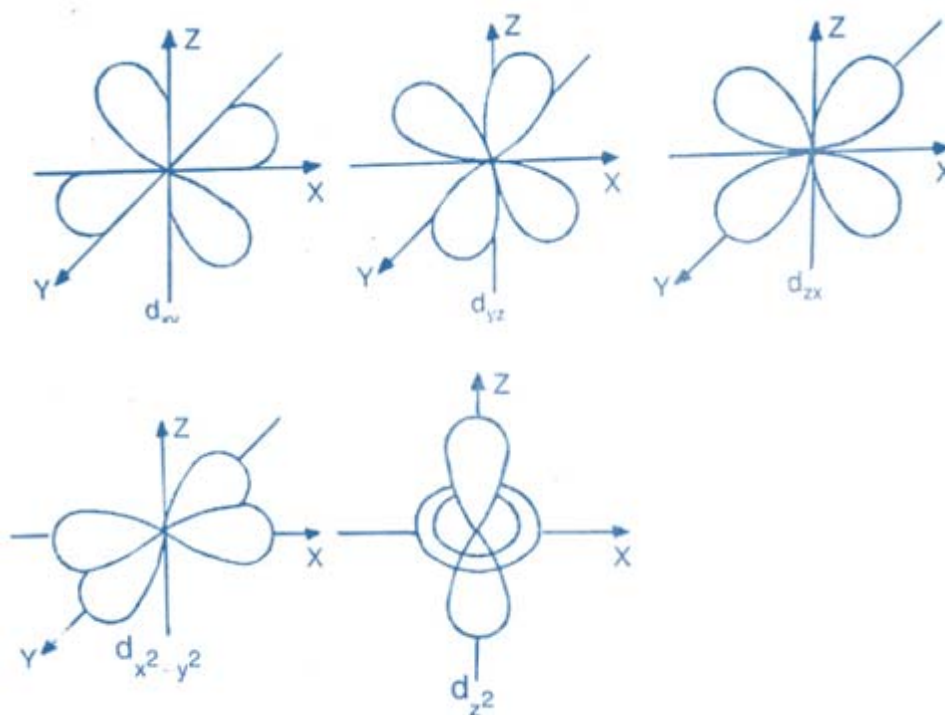
$$\begin{aligned} \Delta G^\circ &= -nFE^\circ_{\text{cell}} \\ &= -(1 \times 96500 \times 0.14) = -13.51 \text{ kJ mol}^{-1} \end{aligned}$$

(iii) Iron(II) ion forms octahedral complexes, the orbitals are split into two energy levels. Using the Cartesian axes shown below, draw separate labelled diagrams of

1. one of the d orbitals at the **lower** energy level in an octahedral complex,
2. one of the d orbitals at the **higher** energy level in an octahedral complex.



[2]



(iii) Using your diagrams in (b)(ii), explain why the d subshell of a transition metal ion is split into two energy levels in an octahedral complex.

- The $d_{x^2-y^2}$ orbital and the d_{z^2} orbitals have their lobes pointing at the ligands along the x and y axes, and z axis respectively, hence they experience greater repulsion from the ligands. On the other hand, the d_{xy} , d_{xz} and d_{yz} orbitals experienced less repulsion as their lobes are in between the coordinate axes.
- As a result, the five d orbitals are split into 2 energy levels, with the $d_{x^2-y^2}$ and d_{z^2} orbitals having the higher energy level and the d_{xy} , d_{xz} and d_{yz} orbitals having the lower energy level.

Most transition element complexes are coloured.

(iv) Explain why ferrocene is orange in colour.

[2]

Fe^{2+} ion has partially filled d orbitals. In the presence of C_5H_5^- ligands, the d orbitals are split into two groups with an energy gap, ΔE . This effect is known as d orbital splitting.

During the $d-d$ transition, the d electrons from the lower energy d orbitals absorb a blue wavelength of light from the visible spectrum and gets promoted to a *higher* energy d orbital. The orange colour observed in ferrocene is complementary of the blue colour absorbed.

(v) Aqueous Fe^{2+} ion is green in colour, suggest and explain if water causes a larger or smaller split between the two groups of $3d$ orbitals as compared to C_5H_5^- .

[2]

With H_2O ligands, the green $\text{Fe}^{2+}(\text{aq})$ ion absorbed red which has a longer wavelength and lower ΔE than the blue absorbed when C_5H_5^- is the ligand.

Hence, H_2O ligands causes a smaller splitting of the $3d$ orbitals of Fe^{2+} .

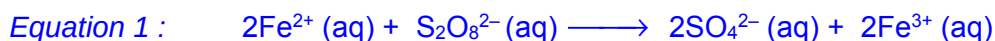
(vi) The reaction between $\text{S}_2\text{O}_8^{2-}$ and I^- is slow in the absence of catalyst. $\text{Fe}^{2+}(\text{aq})$ can be used as catalyst for the reaction between $\text{S}_2\text{O}_8^{2-}$ and I^- .

Explain with the aid of equations how it works.

[3]

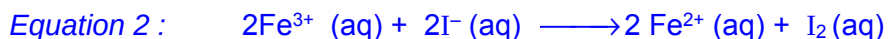
With $\text{Fe}^{2+}(\text{aq})$ catalyst:

Step 1: Fe^{2+} intermediate reacts with $\text{S}_2\text{O}_8^{2-}$



$$E^\circ_{\text{cell}} = 2.01 - 0.77 = +1.24 \text{ V} > 0 \Rightarrow \text{reaction is spontaneous}$$

Step 2: Fe^{3+} reacts with I^- .

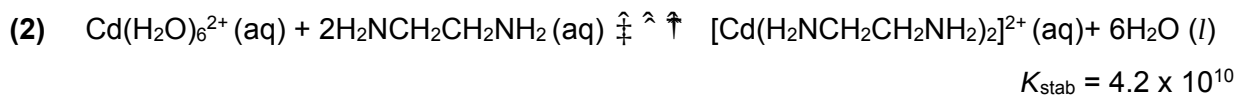
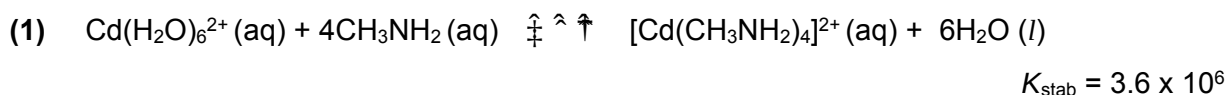


$$E^\circ_{\text{cell}} = 0.77 - 0.54 = +0.23 \text{ V} > 0 \Rightarrow \text{reaction is spontaneous}$$



Both steps are spontaneous since oppositely charged ions are involved and attract each other. Activation energy is lowered and thus reaction is faster (or kinetically feasible).

(c) Cadmium ions form complexes with primary amines and with 1,2-diaminoethane.



K_{stab} is an equilibrium constant that accounts for the formation of a complex. It is a measure of the strength of the interactions between the ligands and the metal centre that come together to form the complex.

The values for ΔH° and ΔG° for equilibria (1) and (2), and the value of ΔS° for equilibrium (1), are given in the table below. All values are determined at a temperature of 298 K.

Equilibrium	$\Delta H^\circ / \text{kJ mol}^{-1}$	$\Delta G^\circ / \text{kJ mol}^{-1}$	$\Delta S^\circ / \text{J mol}^{-1} \text{K}^{-1}$
(1)	- 57.3	- 37.4	- 66.8
(2)	- 56.5	- 60.7	-

(i) Suggest a reason why the ΔH° values for the two equilibria are very similar.

[1]

Each complex is formed by making four N–Cd bonds and breaking six O–Cd bonds
 OR
 same types of / similar bonds forming / breaking
 OR
 same number of bonds forming / breaking

(ii) Calculate the standard entropy change of reaction, ΔS° in equilibrium (2).

[2]

$$\begin{aligned} \Delta S &= (\Delta H - \Delta G) / T \\ &= (60.7 - 56.5) \times 1000 / 298 \\ &= +14.1 \text{ J mol}^{-1} \text{K}^{-1} \end{aligned}$$

(iii) Suggest a reason for the difference between the ΔS° you have calculated for equilibrium (2) and that for equilibrium (1) given in the table.

[1]

For equilibrium (1), 5 moles of reactants react to form 7 moles of products while equilibrium (2), 3 moles of reactants react to form 7 moles of products. Hence, there is more disorder in equilibrium (2) than equilibrium (1) and hence ΔS° for equilibrium (2) is more positive.

(iv) Which of the two complexes is more stable? Give a reason for your answer.

[1]

The $[\text{Cd}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_2]^{2+}$ in equilibrium 2 is more stable because either K_{stab} is larger in magnitude or ΔG° is more negative.

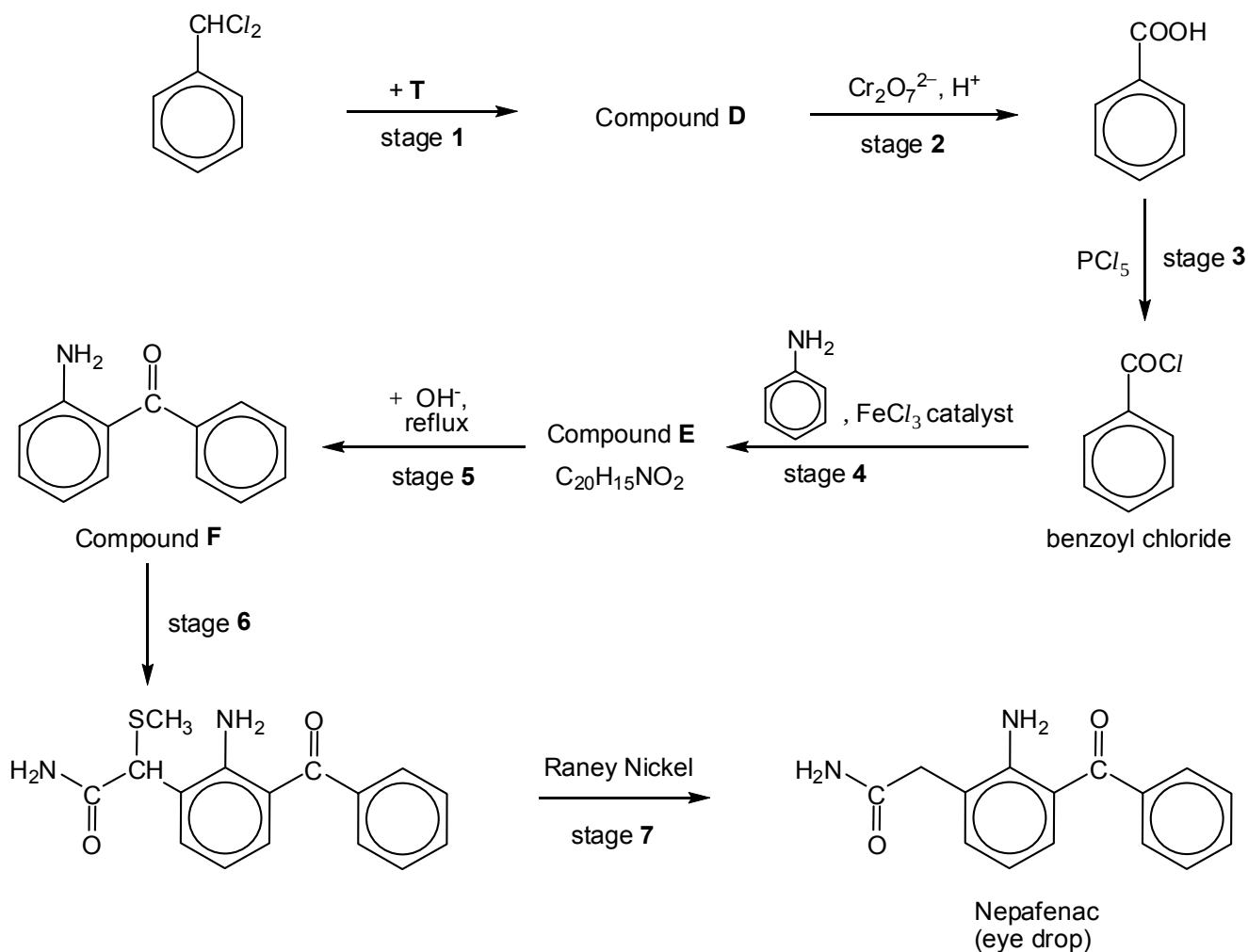
[Total: 23]

Section B

Answer **one** question from this section.

4(a) *Nepafenac* is a prescription eye drop that is used to treat pain and inflammation associated with eye surgery.

The following scheme shows a synthesis of *Nepafenac*.



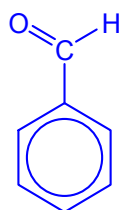
(i) State the type of reaction occurring in stage 3 and stage 7.

[2]

Stage 3: Nucleophilic acyl substitution

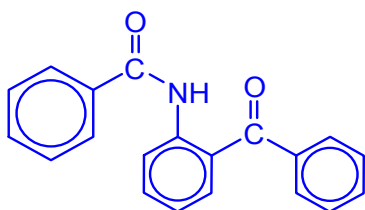
Stage 7: Reduction

- (ii) Draw the structure of compound **D**. Suggest the reagent and conditions required in stage 1. [2]



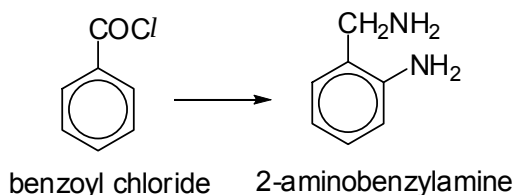
NaOH (aq), heat

- (iii) In stage 4, two reactions are involved. Draw the structure of compound **E** and state the reactions that occurred. [2]



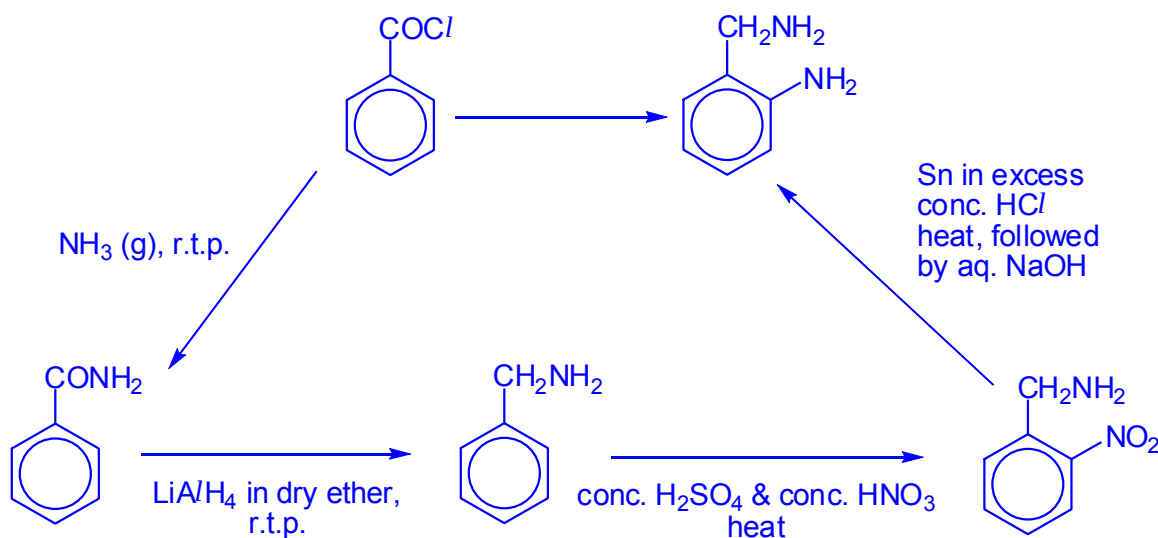
Electrophilic substitution and Condensation / Nucleophilic acyl substitution

- (iv) Benzoyl chloride can be converted into 2-aminobenzylamine, an important starting material in the synthesis of quinazoline, a source of useful pharmacophores for new drug development.

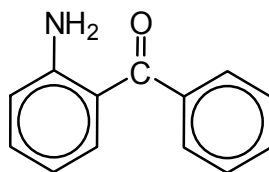


Suggest a 4-step synthesis of 2-aminobenzylamine from benzoyl chloride. You should state the reagents and conditions needed for each step, and show clearly the structures of any intermediate compounds.

[4]



- (v) A student made a statement regarding a reaction with compound **F**: “Cyanide, CN^- , is acting as a **Bronsted–Lowry base** to attack the carbonyl group of compound **F** to form a cyanohydrin.”

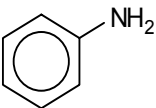
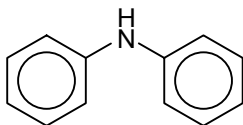
Compound **F**

Based on your understanding of a *Bronsted–Lowry* base, explain why the student's statement is wrong.

[1]

A *Bronsted–Lowry* base is a proton acceptor. In this reaction, the cyanide is behaving as a Lewis base that act as an electron–pair donor.

- (vi) Data about phenylamine and diphenylamine are given in the table below.

Compound	 phenylamine	 diphenylamine
pK_b value	9.39	13.21

Give **two** reasons, why pK_b of phenylamine is lower than that of diphenylamine.

[2]

Phenylamine is a stronger base than diphenylamine. There is greater delocalisation of lone pair of electrons on N atom into the two benzene ring of diphenylamine as compare to phenylamine.

There is also greater steric hindrance of the 2 R group of diphenylamine.
The lone pair on N atom of diphenylamine is less available to accept a proton.

- (b) Compound **J**, $C_{10}H_8O_3$, does not decolourise bromine water nor give effervescence with magnesium strip. **J** gives a silver mirror with Tollens' reagent but does not give a precipitate with hot alkaline Cu^{2+} solution.

On prolonged heating with acidified $KMnO_4$, **J** gives benzene-1,2,3-tricarboxylic acid as the only organic product. When heated with dilute H_2SO_4 , **J** forms compound **K**, $C_{10}H_{10}O_4$. **K** gives a yellow precipitate with alkaline aqueous iodine.

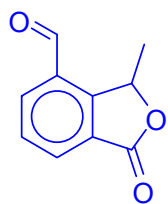
J reacts with $LiAlH_4$ to form compound **L** (with an internal plane of symmetry), $C_{10}H_{14}O_3$. However, **J** forms compound **M**, $C_{10}H_{10}O_3$ with $NaBH_4$.

All compounds are optically active but only **K** reacts with sodium carbonate.

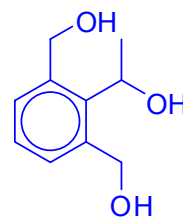
Suggest the structure for **J**, **K**, **L** and **M** and explain the reactions described.

[7]

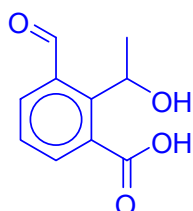
- **J** has comparable number of C and H atoms and no. of carbon atoms > 6
 $\rightarrow$ **J** contains a benzene ring.
- **J** does not undergo electrophilic substitution (or electrophilic addition) with aqueous bromine. **J** does not contain phenol (or alkene group).
- **J** does not undergo acid metal displacement with Mg. **J** does not contain alcohol/phenol/carboxylic acid group.
- **J** undergoes oxidation with Tollens' reagent but does not undergoes oxidation with Fehling's reagent. $\rightarrow$ **J** contains an aromatic aldehyde.
- **J** undergoes oxidation with hot acidified $KMnO_4$ to give benzene-1,2,3-tricarboxylic acid as the only organic product. Substituents the on benzene ring are on 1, 2 and 3 position.
- When heated with dilute H_2SO_4 , **J** undergoes acidic hydrolysis to form **K**. Since only one compound is formed / no decrease in no. of carbon atoms.
 - **J** contains cyclic ester.
 - **K** contains carboxylic acid and alcohol group.
- **K** undergoes oxidation with alkaline aqueous iodine. $\rightarrow$ **K** contains $RCHOH(CH_3)$
- **J** undergoes reduction with $LiAlH_4$ to form compound **L** and 6 H atoms are added to **L**.
 - **J** contains aldehyde and cyclic ester.
 - **L** contains 3 alcohol groups
- **J** undergoes reduction with $NaBH_4$ to form compound **L** and only 2 H atoms are added to **M**.
 - **J** contains aldehyde and cyclic ester.
 - **L** contains 1 alcohol group
- **K** undergoes acid-base / neutralisation with sodium carbonate. $\rightarrow$ **K** contains carboxylic acid group.
- All compound contains chiral carbon and does not an internal plane of symmetry.

Compound **J**:Chemical Formula: $C_{10}H_8O_3$

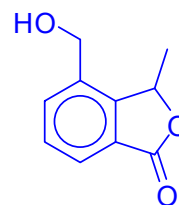
- No phenol, carboxylic acid, alcohol FGs
- Aromatic aldehyde
- Cyclic ester
- Forms benzene-1,2,3-tricarboxylic acid on strong oxidation
- Undergoes reduction to form **L** (gain of 6 H atoms)

Compound **L**:Chemical Formula: $C_{10}H_{14}O_3$

- Product of reduction of **J** (using $LiAlH_4$)
- Reduction of ester in **J** to alcohols in **L** (gain of 4 H)
- Reduction of aldehyde in **J** to primary alcohol in **L** (gain of 2H)
- Internal plane of symmetry

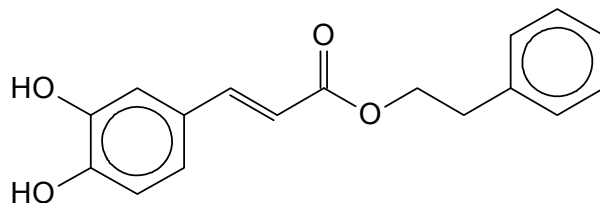
Compound **K**:Chemical Formula: $C_{10}H_{10}O_4$

- Contains carboxylic acid
- Acidic hydrolysis product of **J** (ester linkage)
- Contains a $CH_3CH(OH)-$

Compound **M**:Chemical Formula: $C_{10}H_{10}O_3$

- Product of reduction of **J** (using $NaBH_4$)
- Reduction of aldehyde in **J** to primary alcohol in **L** (gain of 2 H)

- 5 Caffeic acid phenethyl ester (CAPE) is a natural occurring compound, found in propolis from honeybee hives. It is currently being studied for its anti-carcinogenic and anti-inflammatory properties. It is the ester of caffeic acid and 2-phenylethanol.



caffeic acid phenethyl ester (CAPE)

- (a) Till date, there has been two well-established reaction schemes to synthesise CAPE. Study **Fig. 2.1** and **Fig 2.2** carefully and answer the questions that follow.

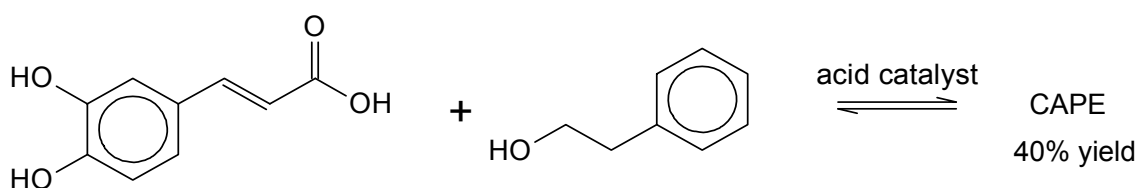


Fig. 2.1

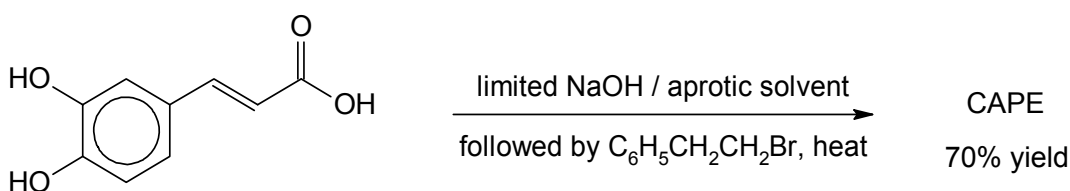


Fig. 2.2

- (i) State the type of reaction occurring in each of the above synthesis method.

[2]

Fig 2.1: Condensation / Nucleophilic Acyl Substitution

Fig 2.2: Nucleophilic substitution

- (ii) To improve the yield of CAPE in **Fig 2.1**, a student suggested using the acyl chloride derivative of caffeic acid instead. Comment on the validity of his suggestion.

[2]

Yes, his suggestion is valid. The reaction between an acyl chloride and an alcohol to form ester is irreversible, hence a higher yield is expected. There are two very electronegative atoms O and Cl bonded to highly electron deficient carbonyl C atom in an acyl chloride. Hence, the carbonyl C atom in an acyl chloride is more susceptible to nucleophilic acyl substitution.

OR

No, his suggestion is invalid. An acyl chloride functional groups can undergo condensation / nucleophilic acyl substitution with the phenol functional groups to form other esters (side products), hence a lower yield is expected.

- (iii) Explain why NaOH has to be added in limited amount rather than in large excess in Fig 2.2. [1]

This is to prevent the halogenoalkane from undergoing nucleophilic substitution with NaOH to form the corresponding alcohols.

- (iv) A protic solvent is one that has labile proton(s) and readily donates its proton(s). Conversely an aprotic solvent does not have any labile proton. Suggest why is it important to use an aprotic solvent in Fig 2.2. [1]

This is to prevent the carboxylate anion from being protonated to form carboxylic acid, hence reducing its ability to function as the nucleophile for nucleophilic substitution with the halogenoalkane.

OR

This is to prevent the generation of phenoxide, another nucleophile which can undergo nucleophilic substitution with $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{Br}$ to form the corresponding ether.

- (v) The reported yield in Fig. 2.2 is only 70%, as there are other by-products. Draw the skeletal formula of one possible by-product, assuming that caffeic acid reacts with $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{Br}$ in a 1:1 ratio. [1]

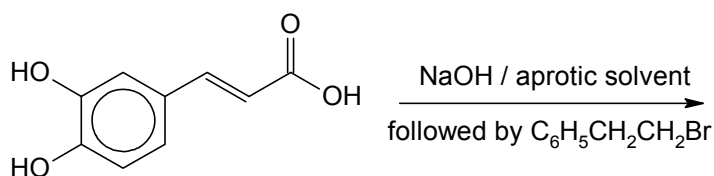
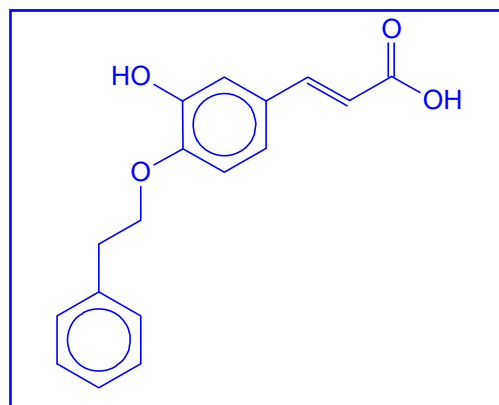
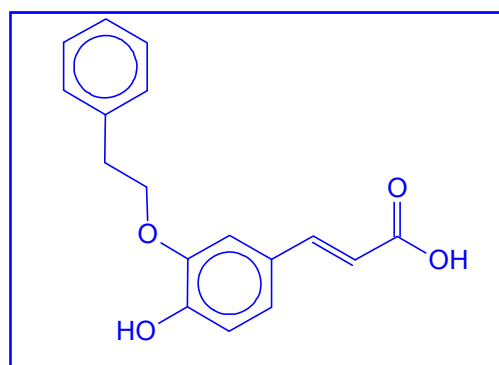


Fig. 2.2

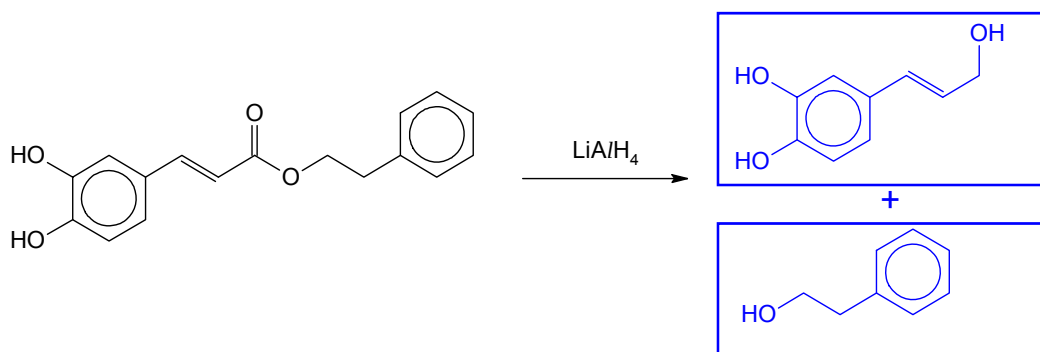


OR

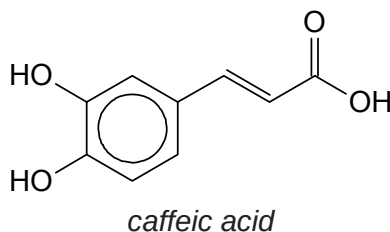


- (vi) Draw the structure of the resultant product(s) when CAPE is reacted with LiAlH_4 .

[1]



- (b) The structure of *caffeic acid* is shown below.



- (i) *Caffeic acid* contains two organic functional groups with pK_a values lower than that of an alcohol. Identify these functional groups.

Compare and account for the relative pK_a values of these two organic groups.

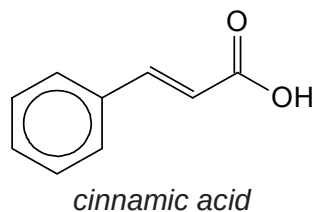
[3]

The carboxylic acid functional group has a lower pK_a value than the phenol functional group. Carboxylic acid is a stronger acid than phenol.

The phenoxide ion is stabilised by charge delocalisation, where the lone pair of electrons on the oxygen atom of the phenoxide ion is delocalised into the benzene ring. This reduces the intensity of the negative charge on the oxygen atom of the phenoxide ion, hence stabilising the phenoxide ion. Phenol is stronger acid than alcohol with a higher tendency to dissociate in aqueous medium to give H^+ .

The carboxylate anion (RCOO^-) is resonance stabilised by the delocalisation of the negative charge over the C atom and both electronegative oxygen atoms in the carboxylate anion. Hence, the carboxylate anion is more stable than phenoxide ion. There is a greater tendency for the carboxylic acid to dissociate in aqueous medium to give H^+ as compared to phenol.

- (ii) Suggest a chemical test that can be carried out to distinguish between a sample of *caffeic acid* and *cinnamic acid*.



[2]

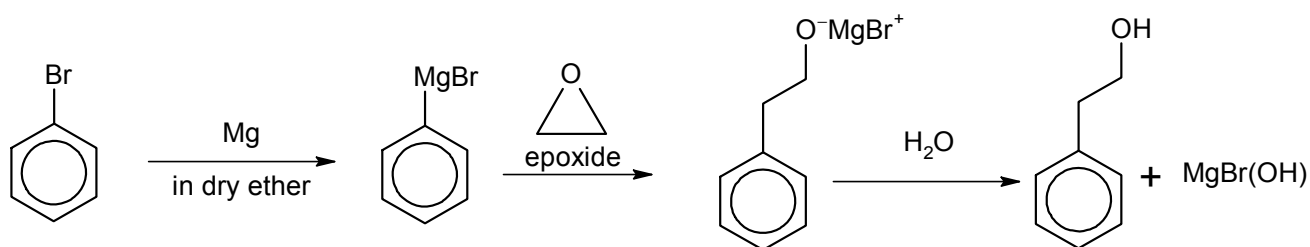
Test: Add aqueous neutral FeCl_3 to each compound separately at r.t.p.

Observations:

Caffeic acid: Violet complex / colouration will be formed.

Cinnamic acid: No violet complex will be formed.

- (c) The synthesis of 2-phenylethanol from bromobenzene is shown below.

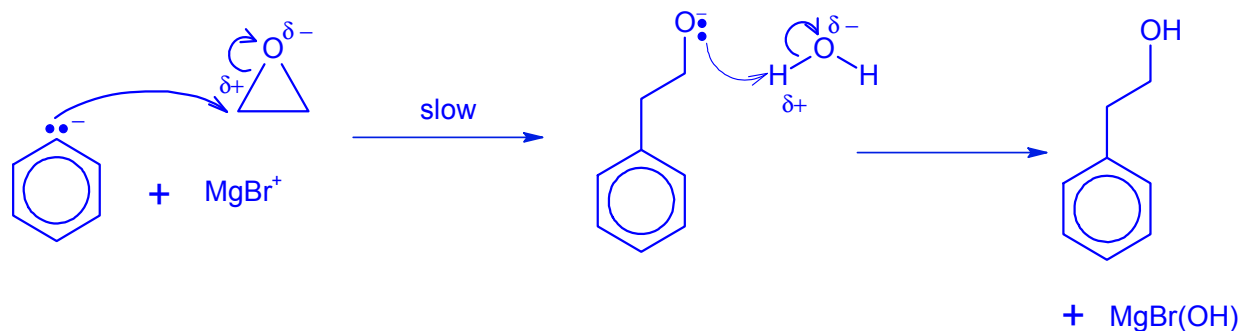


In the above synthesis, a Grignard reagent, $\text{C}_6\text{H}_5\text{MgBr}$ is generated. Once generated, the Grignard reagent behaves as a nucleophile. The C_6H_5 in $\text{C}_6\text{H}_5\text{MgBr}$ behaves like an anion, C_6H_5^- and is a strong Lewis base. The Grignard reagent is useful for the formation of carbon-carbon bonds.

The reaction between the Grignard reagent, $\text{C}_6\text{H}_5\text{MgBr}$ and the epoxide, $\text{C}_2\text{H}_4\text{O}$ is a bimolecular reaction and the intermediate formed will subsequently react with water to form 2-phenylethanol.

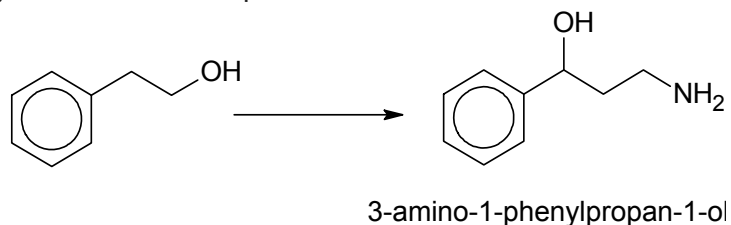
From the information provided, outline the mechanism to account for this reaction. In your answer, show any relevant charges, lone pairs of electrons and movement of electrons by curly arrows.

[2]

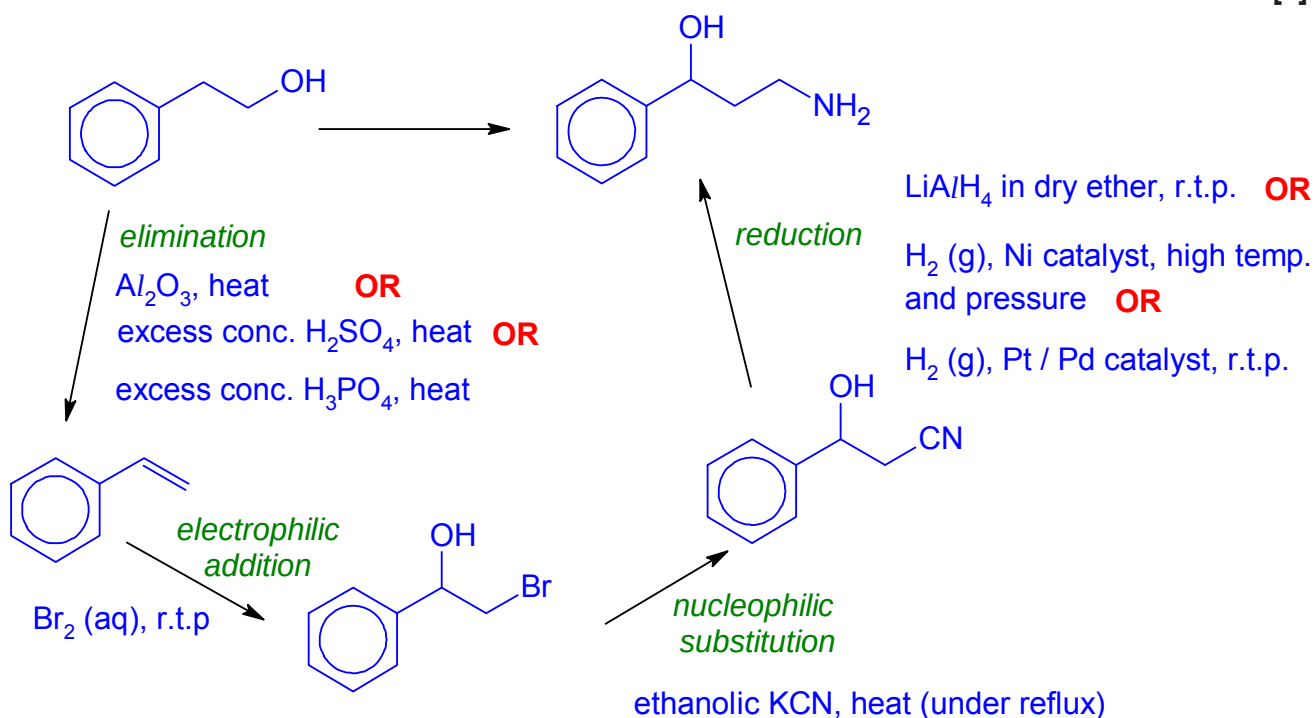


- (d) Suggest a 4-step synthesis of 3-amino-1-phenylpropan-1-ol from 2-phenylethanol.

The synthesis involves an intermediate compound that decolourises aqueous bromine in the dark. You should state the reagents and conditions needed for each step, and show clearly the structures of any intermediate compounds.



[3]



- (e) A 0.460 g sample of gaseous methanoic acid, HCOOH was found to only occupy a volume of 173 cm³ at a temperature of 160°C and a pressure of 1.04×10^5 Pa.

Using the ideal gas equation and the above information, determine the molar mass of this gaseous sample. Hence, draw a possible structure at this temperature.

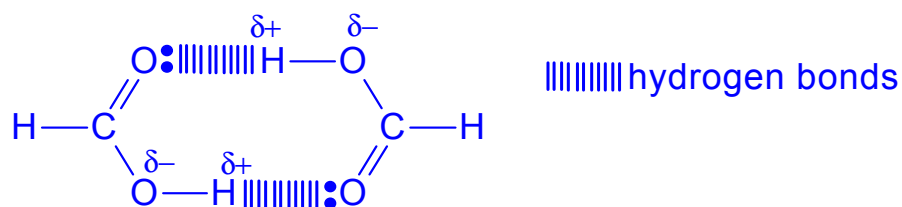
[2]

Applying $pV = nRT$

$$pV = \frac{m}{M} RT$$

$$\text{Molar mass, } M = \frac{mRT}{pV} = \frac{0.460 \times 8.31 \times (160 + 273)}{(1.04 \times 10^5)(173 \times 10^{-6})} = 92.0 \text{ g mol}^{-1}$$

The molar mass of 92.0 g mol⁻¹ is doubled of the expected molar mass of HCOOH (46.0 g mol⁻¹)



[Total: 20]

END OF PAPER